UniComs

CLI Based Communication

Team # 8

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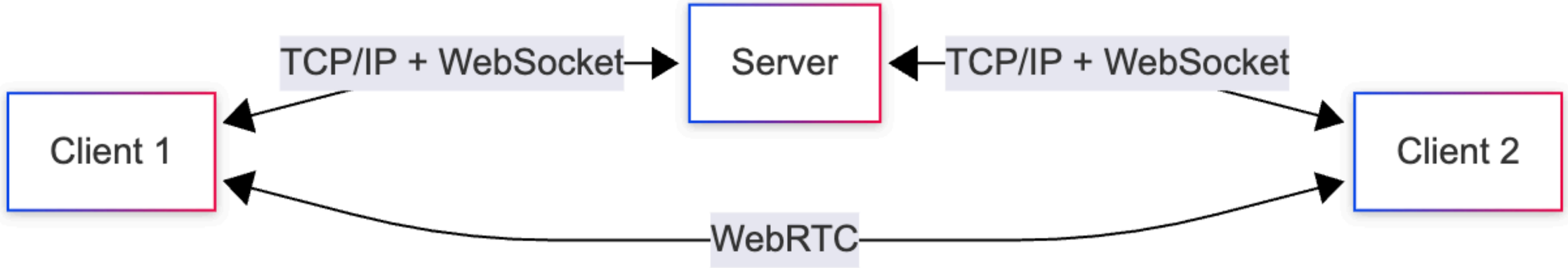
Motivation

- Abstraction free
- Framework-free engineering
- System-call learning
- Security awareness

Project Overview

- Real time chat app
- Completely CLI Based
- Socket Implementation from scratch
  - Client <--> Server
- UDP Voice Streaming using RTP
  - Client <--> Client

Project Architecture



Current Feature List

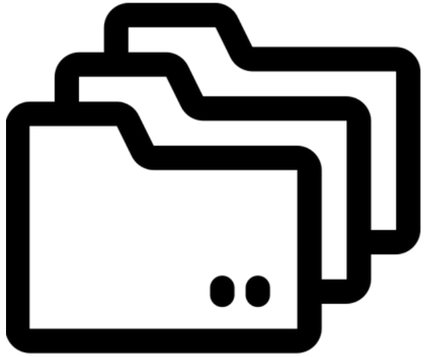
- LAN Based Direct Text Chat
- Login/Signup System
- File transfer protocol
- Persistent Message History

Lan Based Direct Text Chat

→ Socket Implementation
→ Client1->Server->Client2

Login/Signup System

→ Session Based
→ User data in txt



File Transfer Protocol

→ Converting into binary chunks
→ headers with every packet

Server Log

→ Stored in file system
→ All activities are stored

LIVE DEMONSTRATION

Time-Line

Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10
Project planning, finalize architecture									
	Build Server networking core								
		Build Client Core: (Sockets, Parser...)							
			User authentication						
				Implement Direct Messaging					
					Utility Commands				
						File Transfer System			
							Group Chat System		
								Admin Dashboard	
									Integration, Testing and Deployment

Thank **You!**